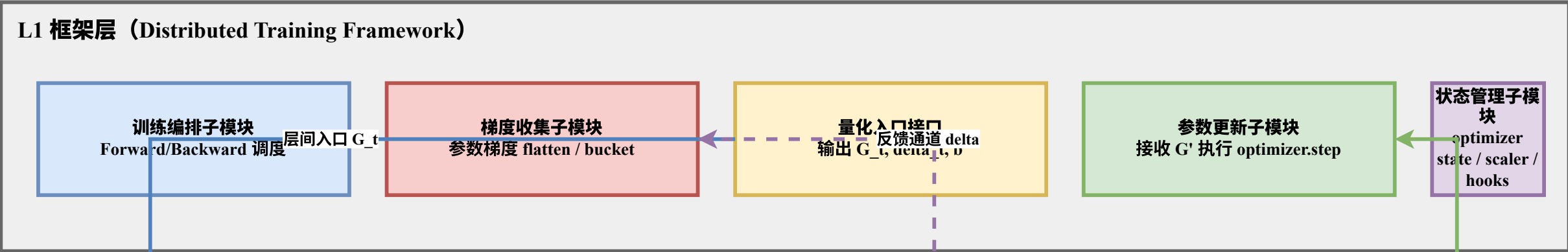
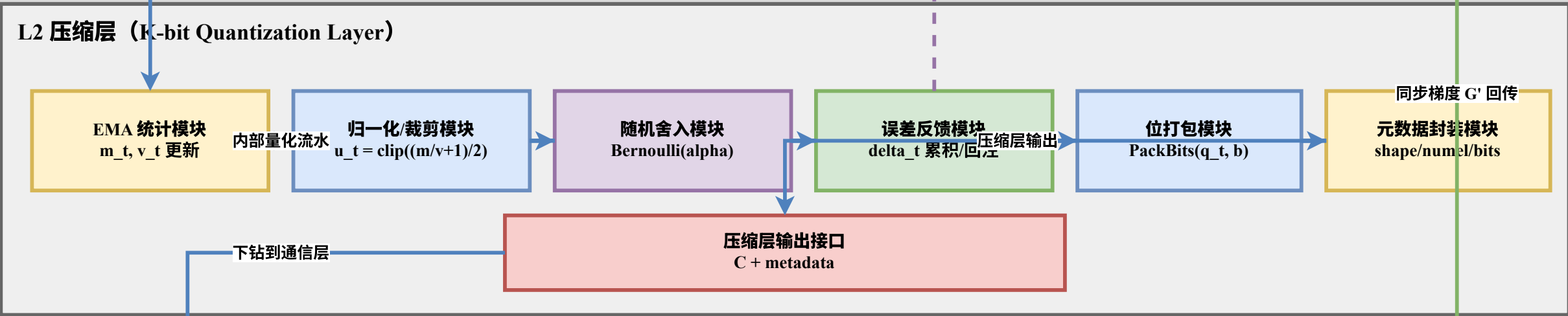


统一 k-bit/1-bit 随机舍入量化在分布式框架中的细粒度分层集成架构

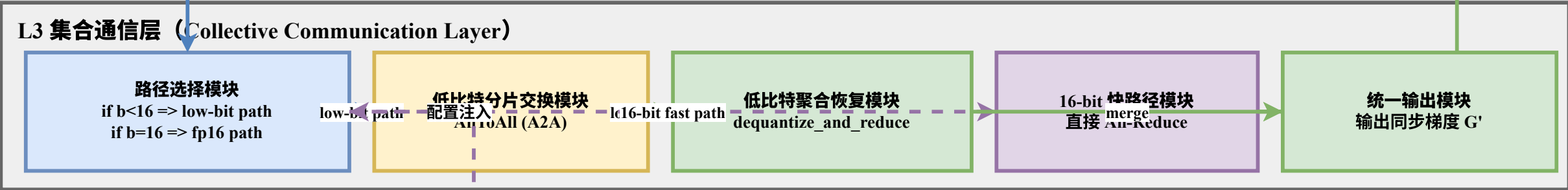
L1 框架层 (Distributed Training Framework)



L2 压缩层 (K-bit Quantization Layer)



L3 集合通信层 (Collective Communication Layer)



全局配置：位宽 b in {1,2,4,8,16}; WorldGroup; 误差反馈开关; 压缩元数据协议